

Nexgile-DecarbX Platform

Enterprise Data Model & Database Architecture Specification
Version 1.0 | Feature Complete Release

1. System Architecture Overview

Nexgile-DecarbX is built on a **Polyglot Hybrid Database Architecture** combining PostgreSQL (Relational Engine) and MongoDB (Unstructured Document Engine). This dual-store strategy ensures strict ACID transactional integrity for carbon calculations and audit logs while providing dynamic schema flexibility for multi-language vendor questionnaires and raw OCR document extractions.

Database Engine	Primary Function	Core Domain Models
PostgreSQL 15+ (SQLAlchemy ORM)	Relational Storage, Multi-Tenant Scoping, Carbon Accounting Ledger, BOM Trees, Finance & Audit Lineage	Organizations, Facilities, ActivityData, EmissionFactors, PCF, CSRD Disclosures, LineageRecord
MongoDB 6+ (Motor / PyMongo)	Document Store, Dynamic Vendor ESG Survey Schemas, Raw Utility PDF Invoice OCR Extractions	questionnaire_templates, supplier_submissions, raw_ocr_extractions

2. Core Domain Schema Breakdown (Modules 0 to 8)

The platform datastore is structured into 9 modular schema packages:

- **Module 0: Platform Foundation:** Multi-Tenant Hierarchy (Orgs, Entities, Facilities), RBAC (Users, Roles, Permissions), Governance (LineageRecord, DataQualityFlag, Workflow Tasks)
- **Module 1: Carbon Accounting:** Scopes 1, 2, and 3 Activity Data Ledger, Emission Factors Library (15+ countries), Calculation Engine
- **Module 2: Product LCA & PCF:** Product Catalog, SKUs, Bill of Materials (BOM) Tree, PCF Engine, Scenario Isolation (@no_actuals_mutation)
- **Module 3: Supplier Scope 3:** Supplier Master, ESG Campaigns, Questionnaire Submissions (MongoDB), Carbon-Weighted Bidding
- **Module 4: AI Analytics & Decarb:** Time-Series Forecasts, 3-Sigma Anomaly Detector, Invoice OCR Extractions, Monte Carlo Simulations, MACC Curve
- **Module 5: Dashboards & Finance:** Executive Aggregations, Operational Drill-Down Tree, Carbon Budgets, Internal Carbon Pricing, Credit Offsets
- **Module 6: Compliance & Disclosure:** CSRD/ESRS Double Materiality, XBRL DataPoint Tags, CBAM Declarations, EU Taxonomy DNSH Checklist
- **Module 7: Data Integrations:** Integration Connections (CSV, REST API, Webhooks), Field Mappings, Sync Runs, Reconciliation Audit Logs
- **Module 8: Platform Hardening:** Global TSVector Search Queries, Scheduled Reports, Lineage Verification Seals, Admin Audit Logs

3. Detailed Entity Relationship & Field Specification

Table: organizations (Organization)

Column Name	Data Type	Constraints	Description & Business Logic
id	VARCHAR(36)	PK, UUID	Unique multi-tenant organization root identifier
name	VARCHAR(255)	NOT NULL	Corporate legal name (e.g. Global Decarb Corporation)
code	VARCHAR(100)	UNIQUE, NOT NULL	Short tenant code (e.g. GLOBAL_DECARB)
country	VARCHAR(100)	NOT NULL	Headquarters country location
currency	VARCHAR(10)	DEFAULT 'USD'	Base reporting currency for carbon financial pricing

Table: facilities (Facility)

Column Name	Data Type	Constraints	Description & Business Logic
id	VARCHAR(36)	PK, UUID	Unique physical site identifier
org_id	VARCHAR(36)	FK -> orgs.id	Parent tenant organization reference
entity_id	VARCHAR(36)	FK -> entities.id	Parent legal entity reference
name	VARCHAR(255)	NOT NULL	Site name (e.g. Texas Clean Tech Plant)
code	VARCHAR(100)	UNIQUE	Facility code (e.g. FAC_TEXAS)
gross_floor_area_sqm	FLOAT	DEFAULT 0.0	Physical area in square meters for intensity metrics
operational_control	BOOLEAN	DEFAULT TRUE	Scope GHG consolidation methodology boundary flag

Table: activity_data (ActivityData)

Column Name	Data Type	Constraints	Description & Business Logic
id	VARCHAR(36)	PK, UUID	Unique activity record identifier
org_id	VARCHAR(36)	FK -> orgs.id	Multi-tenant organization boundary
facility_id	VARCHAR(36)	FK -> facilities.id	Physical facility generating the activity
scope	VARCHAR(50)	NOT NULL	Scope 1, Scope 2, or Scope 3 category designation
category	VARCHAR(100)	NOT NULL	Stationary Combustion, Mobile Fleet, Grid Energy, etc.
activity_type	VARCHAR(100)	NOT NULL	Specific fuel/energy type (Natural Gas, Electricity Grid)
quantity	FLOAT	NOT NULL	Raw input quantity consumed
unit	VARCHAR(50)	NOT NULL	Measurement unit (kWh, m3, liters, kg, USD)

source_type	VARCHAR(50)	NOT NULL	MANUAL_INPUT, BULK_CSV, CONNECTOR_REST_API, OCR_INVOICE
status	VARCHAR(50)	DEFAULT 'APPROVED'	APPROVED, PENDING_REVIEW, REJECTED

Table: emission_factors (EmissionFactor)

Column Name	Data Type	Constraints	Description & Business Logic
id	VARCHAR(36)	PK, UUID	Unique emission factor library entry
name	VARCHAR(255)	NOT NULL	Factor name (e.g. EPA eGRID Texas Subregion ERCT)
scope	VARCHAR(50)	NOT NULL	Scope 1, Scope 2, or Scope 3
co2e_factor	FLOAT	NOT NULL	Numerical emissions multiplier value (e.g. 0.3851)
unit	VARCHAR(50)	NOT NULL	Denominator unit (e.g. kgCO2e/kWh, kgCO2e/m3)
country	VARCHAR(100)	NOT NULL	Geographic country applicability (15+ countries)
version	VARCHAR(50)	NOT NULL	Source edition tag (e.g. EPA 2026.1, DEFRA 2025, IEA 2025)

Table: lineage_records (LineageRecord)

Column Name	Data Type	Constraints	Description & Business Logic
lineage_id	VARCHAR(36)	PK, UUID	Cryptographic audit trail lineage citation identifier
target_entity_type	VARCHAR(100)	NOT NULL	Target model class (ActivityData, Calculation, Disclosure)
target_entity_id	VARCHAR(36)	NOT NULL	Target primary key reference
source	VARCHAR(255)	NOT NULL	Raw input invoice filename or connector system name
methodology	VARCHAR(255)	NOT NULL	GHG Protocol Corporate Standard / ISO 14064
formula	VARCHAR(255)	NOT NULL	Exact mathematical calculation formula representation
factor_version	VARCHAR(50)	NOT NULL	Exact grid emission factor edition tag used
user_id	VARCHAR(36)	NOT NULL	User ID of creator/calculator

Table: lineage_verifications (LineageVerification)

Column Name	Data Type	Constraints	Description & Business Logic
id	VARCHAR(36)	PK, UUID	Auditor verification record primary key
lineage_id	VARCHAR(36)	FK -> lineage_records	Target lineage citation record audited

auditor_user_id	VARCHAR(36)	FK -> users.id	User ID of external auditor
verification_status	VARCHAR(50)	NOT NULL	VERIFIED, FLAGGED, REJECTED
notes	TEXT	NULLABLE	Auditor assurance commentary and verification notes
verified_at	DATETIME	NOT NULL	Timestamp of audit sign-off

Table: products (Product) & boms (BOM)

Column Name	Data Type	Constraints	Description & Business Logic
product.id	VARCHAR(36)	PK, UUID	Product master catalog record
product.code	VARCHAR(100)	UNIQUE	Product SKU/Code (e.g. PROD-SOLAR-X1)
bom.product_id	VARCHAR(36)	FK -> products.id	Parent product reference
bom.parent_bom_id	VARCHAR(36)	Self-FK	Nested sub-assembly hierarchy linkage
bom.component_name	VARCHAR(255)	NOT NULL	Component name (e.g. Aluminum Enclosure Frame)
bom.quantity	FLOAT	NOT NULL	Bill of Materials component quantity
bom.loss_rate_pct	FLOAT	DEFAULT 0.0	Manufacturing scrap rate percentage loss

Table: disclosure_datapoints (DisclosureDataPoint)

Column Name	Data Type	Constraints	Description & Business Logic
id	VARCHAR(36)	PK, UUID	Specific regulatory disclosure requirement data point
disclosure_id	VARCHAR(36)	FK -> disclosures.id	Parent regulatory report (CSRD, CBAM, TCFD)
code	VARCHAR(100)	NOT NULL	Requirement code (e.g. ESRS E1-6 Scope 1 Emissions)
xbrl_tag	VARCHAR(100)	NOT NULL	Regulatory XBRL taxonomy tag for automated filing
lineage_id	VARCHAR(36)	NOT NULL	Primitive LineageRecord citation proving regulatory value

4. Data Security & Multi-Tenant Isolation Rules

- Strict Tenant Scoping:** Every database query enforces `org_id == current_user.org_id` filtering.
- Immutable Lineage Citations:** LineageRecord table entries are append-only. They are never updated or deleted.
- Scenario Isolation (@no_actuals_mutation):** What-if and alternative material simulation runs write exclusively to `scenario_pcf` and `scenario_forecasts`, guaranteeing core ledgers are never mutated.